

Erdős Problem #341

EP-341

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A = \{a_1 < \dots < a_k\}$ be a finite set of positive integers and extend it to an infinite sequence $\bar{A} = \{a_1 < a_2 < \dots\}$ by defining a_{n+1} for $n \geq k$ to be the least integer exceeding a_n which is not of the form $a_i + a_j$ with $i, j \leq n$. Is it true that the sequence of differences $a_{m+1} - a_m$ is eventually periodic?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>